

Project Report

A Deep Learning Approach To prevent Road Accidents Detecting Drivers' Drowsiness

Submitted By

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Abstract

Driver drowsiness is one of the major causes of road accidents worldwide. Detecting fatigue at an early stage can significantly improve road safety and reduce accidents. In this project, a deep learning based driver drowsiness detection system is proposed using facial image frames extracted from videos of multiple individuals. Four models were implemented and compared: Custom CNN, VGG16, CNN+LSTM, and CNN+Bidirectional LSTM. A person-wise train, validation, and test split was used to prevent data leakage and ensure fair evaluation. Experimental analysis shows that temporal deep learning models outperform frame-based models, with CNN+LSTM giving the best generalization performance. The proposed system is extended for real-time vehicle safety applications.

1 Introduction

Driver drowsiness is one of the leading causes of road accidents across the world. Long driving hours, lack of sleep, fatigue, and stress often reduce the alertness of drivers, which may result in delayed reactions, poor judgment, and serious collisions. Therefore, an automatic drowsiness detection system is highly important for improving road safety.

Traditional drowsiness detection methods depend on physiological sensors such as Electroencephalogram (EEG), heart rate monitors, or steering behavior analysis. Although such methods can be effective, they are often intrusive, expensive, or inconvenient for daily use. In contrast, computer vision based systems are non-contact, low-cost, and more practical for real-world deployment.

Recent developments in deep learning have significantly improved image understanding and sequential pattern recognition. Convolutional Neural Networks (CNNs) are highly effective in extracting facial features such as eye closure, yawning, and head posture, while Long Short-Term Memory (LSTM) networks can learn temporal changes occurring over time.

In this project, a driver drowsiness detection framework based on deep learning is proposed using facial image frames extracted from videos of multiple individuals. The images are categorized into two classes: drowsy and non-drowsy. To ensure fair evaluation, person-wise train, validation, and test splitting was performed so that the same individual does not appear in multiple sets.

Four different models were implemented and compared:

- Custom Convolutional Neural Network (CNN)
- VGG16 Transfer Learning Model
- CNN + LSTM Hybrid Model
- CNN + Bidirectional LSTM Hybrid Model

The main objective of this work is to determine which architecture provides the best performance for accurate and reliable drowsiness detection. Experimental results indicate that temporal models perform better than frame-based models because they capture motion patterns and gradual facial changes associated with drowsiness.

2 Literature Review

Driver drowsiness detection has attracted significant research attention due to its direct impact on road safety. Over the years, different approaches have been proposed ranging from traditional machine learning methods to modern deep learning based systems.

Early drowsiness detection systems mainly relied on handcrafted visual features such as Eye Aspect Ratio (EAR), blink frequency, mouth opening ratio, and head pose estimation. These features were extracted manually from facial landmarks and then classified using machine learning algorithms such as Support Vector Machines (SVM), Random Forest, and Decision Trees. Although these methods are computationally efficient, their performance depends heavily on feature engineering and controlled lighting conditions.

Some researchers also used physiological signals such as Electroencephalogram (EEG), Electrocardiogram (ECG), and heart rate variability for fatigue detection. These methods often provide high accuracy, but they require wearable sensors attached to the driver, making them uncomfortable and less practical for real-time use.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) became widely used for driver monitoring systems. CNNs automatically learn important spatial features from facial images such as eye closure, yawning, and facial expressions. Models such as AlexNet, VGG16, ResNet, and MobileNet have shown promising performance in image classification tasks.

However, CNN based methods process images independently and fail to capture temporal dependencies between consecutive frames. Since drowsi-

ness is a gradual process rather than a sudden event, sequence modeling becomes important.

To overcome this limitation, hybrid models combining CNN with Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM) were introduced. In these models, CNN extracts spatial features from each frame, while LSTM learns temporal changes such as slow eye closure and repetitive blinking over time.

Compared to existing methods, the proposed work performs a comprehensive comparison among four architectures: Custom CNN, VGG16, CNN+LSTM, and CNN+Bidirectional LSTM using person-wise dataset splitting. This strict subject-independent evaluation reduces data leakage and provides a more realistic estimate of real-world performance.

Thus, this project focuses not only on classification accuracy but also on model generalization for unseen individuals, which is essential for practical driver drowsiness detection systems.

3 Proposed Methodology

The proposed system aims to automatically detect whether a driver is in a **drowsy** or **non-drowsy** state using facial image frames extracted from videos. A complete deep learning pipeline was developed consisting of dataset preparation, preprocessing, person-wise data splitting, model training, and performance evaluation.

3.1 Overall Workflow

The complete workflow of the proposed system is summarized below:

1. Collect image frames extracted from driver videos.
2. Organize frames into two classes: drowsy and non-drowsy.
3. Perform person-wise train, validation, and test split.
4. Resize and normalize all images.
5. Generate frame sequences for temporal models.
6. Train four deep learning models.
7. Evaluate using standard performance metrics.

3.2 Dataset Organization

- **Dataset:**DDD(updated 4 years ago)
- **link:**<https://www.kaggle.com/datasets/ismailnasri20/driver-drowsiness-dataset-ddd>

The dataset consists of facial image frames extracted from videos of 28 individuals. Two folders were used:

- **Drowsy:** 22348 images
- **Non-Drowsy:**19445 images

Each image file contains a subject prefix and frame number. Examples:

- A001.png, A002.png, A003.png
- a001.png, a002.png, a003.png

Here:

- Uppercase prefixes represent drowsy images.
- Lowercase prefixes represent non-drowsy images.

Thus, the same individual appears in both alert and drowsy conditions.

3.3 Person-wise Data Splitting

To prevent data leakage, the dataset was divided **person-wise** rather than image-wise.

This means:

- If person A is used in training, then person a is also assigned to training only.
- The same subject never appears in train and test simultaneously.

The dataset was divided into:

- Training Set : 19 persons
- Validation Set : 4 persons
- Test Set : 5 persons

This ensures unbiased evaluation on unseen individuals.

3.4 Image Preprocessing

All input images were preprocessed before training.

The preprocessing steps were:

- Resize images to $128 \times 128 \times 3$
- Convert pixel values to floating point
- Normalize pixel values to range $[0, 1]$

This improves numerical stability and faster convergence during training.

3.5 Sequence Generation for Temporal Models

For CNN+LSTM and CNN+BiLSTM models, single images are not sufficient because drowsiness develops over time.

Therefore, image sequences were generated using a sliding window method.

- Sequence Length = 10 consecutive frames
- Sliding Window Step = 5

Example:

$$(F_1, F_2, \dots, F_{10}), (F_6, F_7, \dots, F_{15})$$

Sequences were created separately for each person to preserve temporal continuity. Total no. of sequences are:

Train	Test	Validation
4809	1929	1480

3.6 Models Implemented

Four deep learning architectures were implemented.

3.6.1 1. Custom CNN Architecture

The Custom CNN model was developed for frame-wise binary image classification. It consists of three convolution blocks followed by fully connected layers.

- Input Size: $128 \times 128 \times 3$
- Convolution Layer: 32 filters of size 3×3 , ReLU activation
- MaxPooling Layer: 2×2
- Batch Normalization
- Convolution Layer: 64 filters of size 3×3 , ReLU activation
- MaxPooling Layer: 2×2
- Batch Normalization
- Convolution Layer: 128 filters of size 3×3 , ReLU activation
- MaxPooling Layer: 2×2
- Batch Normalization
- Flatten Layer
- Dense Layer: 128 neurons, ReLU activation
- Dropout: 0.5
- Output Layer: 1 neuron, Sigmoid activation

This architecture learns hierarchical spatial features such as eye closure, facial fatigue, and head posture.

3.6.2 2. VGG16 Transfer Learning Architecture

To improve feature extraction performance, the pretrained VGG16 model was used as the convolutional backbone.

- Base Model: VGG16 pretrained on ImageNet
- Flatten Layer
- Dense Layer: 256 neurons, ReLU activation

- Batch Normalization
- Dropout: 0.5
- Dense Layer: 128 neurons, ReLU activation
- Dropout: 0.3
- Output Layer: 1 neuron, Sigmoid activation

Transfer learning helps leverage previously learned visual representations and improves training efficiency on limited datasets.

3.6.3 3. CNN + LSTM Architecture

The CNN+LSTM model was designed to capture both spatial and temporal information from image sequences.

- Input Shape: (SEQ_LEN , 128, 128, 3)
- TimeDistributed Conv2D: 32 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Batch Normalization
- TimeDistributed Conv2D: 64 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Batch Normalization
- TimeDistributed Conv2D: 128 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Batch Normalization
- TimeDistributed Flatten
- LSTM Layer: 128 units
- Dense Layer: 64 neurons, ReLU
- Dropout: 0.5
- Output Layer: 1 neuron, Sigmoid activation

In this architecture, CNN extracts frame-wise features while LSTM captures temporal dependencies across consecutive frames.

3.6.4 4. CNN + Bidirectional LSTM Architecture

To capture temporal dependencies in both directions, Bidirectional LSTM was used.

- Input Shape: (SEQ_LEN , 128, 128, 3)
- TimeDistributed Conv2D: 32 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Conv2D: 64 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Conv2D: 128 filters, ReLU
- TimeDistributed MaxPooling
- TimeDistributed Flatten
- Bidirectional LSTM: 32 units
- Dropout: 0.6
- Output Layer: 1 neuron, Sigmoid activation

The Bidirectional LSTM processes sequence information in forward and backward directions, allowing richer temporal context learning.

3.7 Training Strategy

All models were trained using:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Batch Size: it varies in different model
- Epochs: 25(custom-CNN,CNN+bi-Lstm),20(CNN+Lstm,VGG)

To reduce overfitting:

- EarlyStopping
- ReduceLROnPlateau
- ModelCheckpoint
- Dropout Regularization

3.8 Performance Evaluation

The trained models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC Score
- Confusion Matrix

For temporal models, person-wise sequential testing was also performed using timeline probability plots.

3.9 Expected Outcome

The system is expected to identify drowsy behavior accurately and robustly across unseen individuals. Temporal models such as CNN+LSTM are expected to outperform image-only models because they capture gradual behavioral changes over time.

4 Experimental Results

This section presents the performance evaluation of the four implemented deep learning models: Custom CNN, VGG16, CNN+LSTM, and CNN+Bidirectional LSTM. All models were trained using the same person-wise train, validation, and test split to ensure a fair comparison.

4.1 Experimental Setup

4.2 Accuracy Comparison of Models

4.3 Additional Performance Metrics

4.4 ROC-AUC Comparison

4.5 Observations

Based on the experimental results, the following observations were made:

- The Custom CNN model achieved high training accuracy but lower validation and test accuracy, indicating overfitting.
- The VGG16 transfer learning model improved generalization performance due to pretrained feature extraction.
- CNN + LSTM performed better than frame-based models because it captured temporal information across consecutive frames.
- CNN + LSTM achieved the best balance between training and test accuracy.
- CNN + Bidirectional LSTM achieved very high training accuracy but comparatively lower test accuracy, showing overfitting due to increased model complexity.
- Temporal models were more effective than image-only models for drowsiness detection.

There are some another person-wise specific observations regarding cnn+lstm model i.e "Time Line probability plot".

- This graph shows the **drowsiness probability over time** for an individual test subject.
- The **X-axis** represents no. of sequences arranged in chronological order.
- The **Y-axis** represents the predicted drowsiness probability ranging from **0 to 1**.
- Each point in the graph corresponds to one sequence prediction generated by the **CNN+LSTM model**.
- Probability values greater than **0.5** indicate a non-drowsy state, while values below 0.5 indicate a drowsy state.
- This visualization helps analyze the **temporal evolution of driver alertness** over time.

4.6 Best Model

From the overall comparison, the **CNN + LSTM** model was selected as the best performing model because it achieved the highest test accuracy and better generalization on unseen individuals.

Table 1: Training, Validation and Test Accuracy Comparison

Model	Validation Accuracy (%)	Test Accuracy (%)
Custom CNN	66.80	58.11
VGG16	54.74	55.19
CNN + LSTM	80.07	75.53
CNN + BiLSTM	76.55	56.66

Table 2: Precision, Recall and F1-Score Comparison

Model	Precision	Recall	F1-Score
Custom CNN	0.57	0.62	0.6
VGG16	0.53	0.74	0.62
CNN + LSTM	0.6977	0.8921	0.7830
CNN + BiLSTM	0.5425	0.7947	0.6448

Table 3: ROC-AUC Score Comparison

Model	ROC-AUC Score
Custom CNN	0.503
VGG16	0.557
CNN + LSTM	0.8472
CNN + BiLSTM	0.5305

Table 4: Overall Ranking of Models

Rank	Model
1	CNN + LSTM
2	VGG16
3	CNN + BiLSTM
4	Custom CNN

5 Summary

This project developed a driver drowsiness detection system using deep learning techniques. Four models were implemented and compared. Frame-based models were effective for static feature extraction, while sequence-based models captured temporal behavior more accurately.

Among all models, CNN+LSTM showed the best balance between performance and generalization. The proposed system is tested in real-time using dashboard cameras for accident prevention.

Future work includes:

- Attention mechanisms
- Transformer-based sequence models
- Larger multi-driver datasets

6 References

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